Muhammet Celal Gokalp

Test & Verification Engineer | Systems Integration

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PROFESSIONAL SUMMARY

Test and verification engineer with nearly 8 years in automotive, working on safety-related driver assistance and automated driving systems. Experienced across the full test lifecycle: writing requirements and test specifications, building test scenarios and test data, running integration, system, and regression tests on simulation, HiL/ViL rigs and real vehicles, and tracing results back to requirements. Comfortable in both V-model (waterfall) and agile teams. I write Python to automate tests and build small test tools, and I pick up new tools quickly. British national, UK resident for 8+ years, and willing to undergo SC clearance.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – Present

FiveAI / Bosch

Cambridge, United Kingdom

- **Test design:** Write test scenarios in OpenSCENARIO 2 for an L4 automated driving stack, from functional to logical to concrete. Built 120+ parameterised scenarios that generate thousands of test cases, including Euro NCAP junction tests and UN R79 lane change compliance tests.
- **Test execution & regression:** Run simulation test suites through the test pipeline (dtflow), check results against pass/fail and KPI criteria, and raise defects with the development teams.
- **Performance testing:** Evaluated performance KPIs, such as comfort limits for acceleration and deceleration, across 1,000+ hours of test data, and reported violations to the development teams.
- **Test automation & tools:** Write Python scripts to validate the stack on both simulation and real-world drive data. Built internal test tools, such as a GUI for JSON test parameters and a local uploader, using AI-assisted coding, and raised pull requests to the main code base.
- **Test data & reporting:** Built data pipelines that move simulation results into front-end dashboards so stakeholders can see test status and performance.
- **Test standards:** On the ADA project, a partnership with Volkswagen Group where FiveAI supplied the simulation tooling, trained partner engineers to use our toolchain. This grew into the Simulation EcoSystem solutions team, where I was a founding member. Created reusable Euro NCAP test templates and helped triage tool and map integration issues (SDB/XODR).

ADAS Systems Engineer (Feature Owner)

11/2018 – 10/2021

AVL Powertrain UK

Coventry, United Kingdom

- **Requirements & traceability:** Based at Jaguar Land Rover. Broke high-level requirements down into low-level ones in DOORS and MKS Integrity within an ASPICE V-model process, and wrote test cases for each system state, traced back to the requirements.
- **Integration & system test:** Ran verification and validation of ADAS features (Rear Cross Traffic Alert, Blind Spot Monitor, Rear Pre-Crash) on Hardware-in-the-Loop (HiL) and Vehicle-in-the-Loop (ViL) rigs and on the proving ground.
- **Fault-finding:** Used Vector CANoe and CANape to read vehicle network data and find the root cause of failures.
- **Field diagnostics:** At the LEVC (London Taxi) factory, inspected returned faulty e-machines: multimeter checks, downloading data from proprietary loggers and sending it to developers and suppliers for root cause analysis.
- **Defect management:** Logged and tracked defects in Rational Team Concert (RTC/RQM). For example, when an alert triggered late at boundary conditions, I re-ran the tests, analysed the logs, and worked with the supplier to change the sensor field of view while keeping false alerts low.
- **Safety-related work:** Wrote ISO 26262 functional safety requirements for a lane change feature. Trained in ISO 26262 and SOTIF (ISO 21448).

Mechanical Engineer Intern

05/2014 – 08/2014

Temsa Global

Adana, Türkiye

- Calculated power and torque needs for a new electric bus powertrain in MATLAB/Simulink, and supported quality work on the production line.

EDUCATION

MSc in Sustainable Automotive Engineering

10/2017 – 09/2018

Warwick Manufacturing Group, Coventry, UK

- **Grade:** 2:1 (Merit). Dissertation: experimental testing to find the variables of a thermal model of an electric motor stator (MATLAB, Simulink, MotorCAD).

BSc in Mechanical Engineering

08/2011 – 02/2017

Ege University, Izmir, Türkiye

- Dissertation: designed, built and tested a 6-DOF Stewart motion platform (SolidWorks, MATLAB, Arduino).

SKILLS

Testing	Test specifications, test scripts and scenarios, integration / system / regression testing, performance KPI testing, verification & validation, acceptance criteria, HiL / ViL / SIL
Requirements	DOORS, MKS Integrity, requirements traceability, ASPICE V-model, ISO 26262 and SOTIF (trained)
Automation & Tools	Python, OpenSCENARIO 2, test pipelines, Git / GitHub, Vector CANoe / CANape, MATLAB / Simulink
Delivery	Agile (Jira, Confluence), V-model / waterfall, defect management (RTC / RQM)
Other	English (C2), Turkish (Native); British national, UK resident 8+ years, willing to undergo SC clearance